

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 6: Distress & Urgency Procedures

Syllabus Source: CAE Oxford (pp. 143-168) | Pass Target: >= 90% AviationExam

1. DISTRESS VS URGENCY STATES (ICAO DEFINITIONS)

Emergency State	Radiotelephony Callout	ICAO Definition & Aeronautical Scenarios
Distress (MAYDAY)	Spoken 3 TIMES: 'MAYDAY, MAYDAY, MAYDAY'	Condition of being threatened by grave and/or imminent danger and of requiring immediate assistance (e.g. dual engine failure, uncontrolled in-flight fire, structural failure, forced landing / ditching).
Urgency (PAN PAN)	Spoken 3 TIMES: 'PAN PAN, PAN PAN, PAN PAN'	Condition concerning the safety of an aircraft or other vehicle, or of some person on board or within sight, but which does NOT require immediate assistance (e.g. critical passenger medical condition, lost orientation with safe fuel, single engine failure in twin cruise).

2. INTERNATIONAL EMERGENCY FREQUENCIES & SQUAWK CODES

A distress or urgency call should normally be made on the frequency in current use. If unable, or if no contact is established, transmit on the international emergency frequencies:

- 121.500 MHz: International VHF Aeronautical Emergency Frequency (VHF Guard). Continuously monitored by military, civil ATC, and oceanic stations.
- 243.000 MHz: International UHF Military Emergency Frequency (UHF Guard).
- 406.0 MHz: Digital Emergency Locator Transmitter (ELT) satellite frequency. Detected by the Cospas-Sarsat constellation; transmits unique 24-bit aircraft hex code. 121.5 MHz is retained for localized homing.

[!] EXAM TRAP (AVIATIONEXAM) - The Three Master Emergency Transponder Squawks (Memorize 100%)
- Squawk 7700: GENERAL EMERGENCY (Distress or Urgency condition); - Squawk 7600: RADIO COMMUNICATION FAILURE (Complete loss of two-way radio); - Squawk 7500: UNLAWFUL INTERFERENCE (Hijacking / aircraft seizure). Note: In an emergency, select 7700 immediately on the Mode A/C transponder. This triggers automated audio and visual alarms on ATC radar consoles!

3. STRUCTURE OF A DISTRESS MESSAGE (ORDER OF PRIORITY)

Under ICAO Annex 10, a distress message must contain the following information in sequence:

Transmission Order	Exact ICAO Radiotelephony Phrasing
1. Distress Call	'MAYDAY, MAYDAY, MAYDAY'
2. Addressed Station	'MADRID RADAR' (or 'ALL STATIONS')
3. Aircraft Callsign	'IBERIA 456' (spoken 3 times on initial call)

4. Nature of Distress	'ENGINE FIRE ON ENGINE NUMBER ONE'
5. Pilot Intentions	'DESCENDING IMMEDIATELY, DIVERTING TO BARCELONA'
6. Present Position & Level	'POSITION 20 NM SOUTH OF REUS, FLIGHT LEVEL 150, HEADING 040'
7. Additional Information	'142 PERSONS ON BOARD, FUEL ENDURANCE TWO HOURS'

4. RADIO SILENCE, RELAY & EMERGENCY TERMINATION

- Imposing Radio Silence: The aircraft in distress or the controlling station can order all other traffic to cease transmitting: 'STOP TRANSMITTING, MAYDAY' (or 'SILENCE MAYDAY').
- MAYDAY RELAY: Transmitted when an aircraft observes another aircraft in distress that cannot transmit its own message: 'MAYDAY RELAY, MAYDAY RELAY, MAYDAY RELAY'.
- Termination of Distress: When the emergency has ended: Aircraft transmits 'CANCEL DISTRESS'; the controlling station transmits 'DISTRESS TRAFFIC ENDED'.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): Number Pronunciation & Reading

REGLAS MNEMOTÉCNICAS DE EXAMEN: 1. Pronunciación de números enteros: - Centenas redondas: 'HUND-red' (ej. 800 = 'AIT HUND-red'). - Millares redondos: 'TOU-SAND' (ej. 5.000 = 'FIFE TOU-SAND'). - Combinación: 38.300 = 'TREE AIT TOU-SAND TREE HUND-red'. 2. Niveles de vuelo (FL): SIEMPRE dígito a dígito (ej. FL 100 = 'FLIGHT LEVEL WUN ZE-RO ZE-RO'). 3. Escala de legibilidad: Del 1 (ininteligible) al 5 (perfecto). 'Read you 3' significa 'legible con dificultad'.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: How is a condition of 'Distress' distinguished from a condition of 'Urgency' in ICAO radiotelephony?

- [A]: Distress uses PAN PAN; Urgency uses MAYDAY
- [B]: Distress indicates grave and imminent danger requiring immediate assistance; Urgency concerns safety but does not require immediate assistance
- [C]: Distress applies only on the ground
- [D]: Urgency requires immediate ditching

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Distress (MAYDAY) = grave and imminent danger requiring immediate assistance. Urgency (PAN PAN) = safety concern not requiring immediate assistance.

Q2: What is the primary international aeronautical emergency frequency in the VHF band?

- [A]: 118.000 MHz
- [B]: 121.500 MHz
- [C]: 123.450 MHz
- [D]: 243.000 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. 121.500 MHz is the international VHF aeronautical emergency frequency (VHF Guard). 243.000 MHz is UHF military guard.

Q3: Which SSR transponder code must be set immediately when an aircraft encounters an in-flight emergency

- [A]: 7000
- [B]: 7500
- [C]: 7600
- [D]: 7700

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [D]. Squawk 7700 indicates general emergency (distress or urgency). 7600 is radio failure; 7500 is unlawful interference.

Q4: What phraseology is used by an air traffic controller to impose radio silence on a frequency during an active emergency?

- [A]: 'Silence on the frequency'
- [B]: 'Stop transmitting, MAYDAY'
- [C]: 'All aircraft maintain radio silence'
- [D]: 'Standby all stations'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Standard ICAO phraseology to impose radio silence is 'STOP TRANSMITTING, MAYDAY'.

Q5: What is the correct phraseology transmitted by an aircraft to cancel a previously declared PAN PAN urgency?

- [A]: 'Emergency cancelled'
- [B]: 'Cancel PAN PAN, situation resolved'
- [C]: 'Disregard PAN PAN'
- [D]: 'Back to normal'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The pilot transmits 'CANCEL PAN PAN' followed by callsign and current status.

Q6: On what satellite frequency do modern digital Emergency Locator Transmitters (ELTs) transmit encoded digital distress data?

- [A]: 121.5 MHz
- [B]: 243.0 MHz
- [C]: 406.0 MHz
- [D]: 1090 MHz

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [C]. Modern digital ELTs transmit digital distress data on 406.0 MHz to rescue satellites, retaining 121.5 MHz for final acoustic homing.